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The Effect of ADHD Diagnoses on Post-Graduate Education

Over the past 3 decades, ADHD diagnoses in the United States have been climbing steadily. Recent questions have been raised as to how ADHD should be identified and treated. This trend has significant implications for policymaking within classroom environments, as evolving approaches to ADHD diagnosis and treatment may influence educational trajectories and long-term academic outcomes. Specifically, are people diagnosed with ADHD more likely to pursue postgraduate education than the non-diagnosed general population? To assess this question, data on children diagnosed with ADHD and their educational attainment from the UNC Add Health dataset will be examined. I expect to find that ADHD-diagnosed individuals have a higher rate of pursuing postgraduate education than the general population because the structure of higher education lends itself complementarily to how it is theorized that diagnosed individuals learn better than when compared to the structure of primary and secondary education.

The Add Health dataset is a nationally representative longitudinal study of U.S. adolescents that tracks their social, behavioral, and health outcomes. It is conducted through a series of in-home interviews, surveys, and follow-ups with participants, beginning in adolescence and continuing into adulthood, along with data collection from schools, parents, peers, and public records. To date, the data has been collected in five check-ins, or waves. The data used was collected in waves I, IV, and V. These variables were measured by surveys and in-home interviews with the subjects and their caregivers. Data on ADHD diagnoses, highest education level attained, household income, maternal education, gender, age, and race were utilized.

Although multiple regressions were run within various subsets of the data, they all maintained this form:

$$edlvl_i = \alpha + \beta^1 adhd_i + \beta^2 age_i + \beta^3 female_i + \beta^4 hhlogincome_i + \beta^5 matedlevl_i + \sum_{r=1}^{R-1} \delta_r race_{ir} + \epsilon_i$$

This regression was run on the entire dataset, a subset of only subjects who had attended some college, and a subset of only subjects who had graduated from college. The regression considering the entire dataset was also ran once treating the variable $edlvl_i$ as an ordinal. This model is used to analyze the causal effect of ADHD (the treatment) on the highest education level attained. Highest education level is correlated with age, household income, maternal education level, and race, so these groups are controlled for. Furthermore, ADHD is more prevalent in males, so gender is controlled for. Variables pertaining to educational outcomes, such as standardized test scores, are not included as controls, as these variables may be dependent outcomes of an ADHD diagnosis. According to the omitted variable bias calculation, these controls account for a 0.371 positive bias in the short regression.

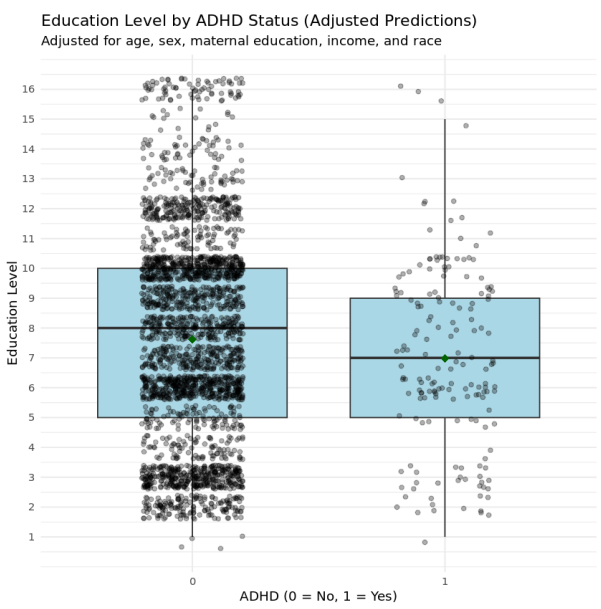
The linear and ordinal regressions run on the entire dataset depict a statistically significant negative relationship between ADHD and education level attained at -0.635 and -0.435 decrease from the intercept of education level 8. This points to a negative correlation between ADHD and education level, with a diagnosis causing a decrease in the highest level attained. In the college subset, the regression shows a statistically significant -0.721 decrease in level attained, from the 8.351 intercept, and the graduate subset resulted in a -0.337 decrease, although this value is not statistically significant. Across these subsets, an ADHD diagnosis can cause a decrease in educational attainment of around half to a full level.

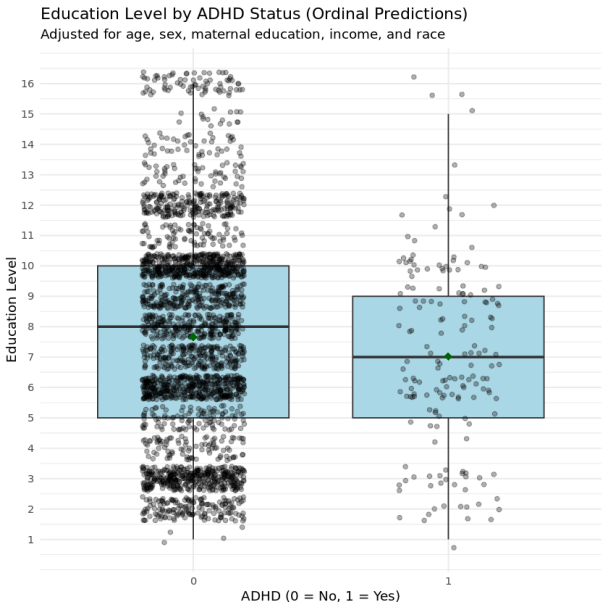
These findings contradict the original hypothesis that individuals diagnosed with ADHD would be more likely to pursue postgraduate education. Instead, the results suggest that an ADHD diagnosis is associated with a statistically significant decrease in highest education level attained, even after controlling for relevant demographic and socioeconomic factors. This suggests that structural elements of higher education may not sufficiently compensate for the challenges ADHD presents, contrary to initial assumptions. From a policy standpoint, these findings highlight the importance of sustained academic support for individuals with ADHD beyond K–12 education. While accommodations often exist in primary and secondary settings, similar support systems in college and graduate school may be less robust or accessible. More longitudinal and qualitative data could help clarify

whether the gap in educational attainment stems from systemic barriers, treatment efficacy, or a combination of both. This may reflect a structural constraint in the education system that acts as a bottleneck, limiting the upward mobility of individuals with ADHD once compulsory schooling ends.

Variable	Measurement Unit	
Highest level of education attained (*edlvl*)	Numeric	2 – some high school or lower 3 – high school diploma 4 – GED 5 – some vocational/technical training (after high school) 6 – some community college 7 – completed vocational/technical training (after high school) 8 – associate or junior college degree 9 – some college 10 – completed college (bachelor's degree) 11 – some graduate school 12 – completed a master's degree 13 – some graduate training beyond a master's degree 14 – completed a doctoral degree 15 – some postbaccalaureate professional education (such as law school, medical school, nursing) 16 – completed a postbaccalaureate professional degree (such as law, medicine, nursing)
Maternal education level (*matedlvl)*		
ADHD diagnosis (*adhd*)	Binary	0 – nondiagnosed 1 – diagnosed
Age (*age)*	Numeric	
Biological sex (*female*)	Binary	0 – male 1 – female
Household income as an adolescent (*hhlogincome*)	Numeric	$0 – $999,999
Race (*race*)	Binary	0/1 – white, black, hispanic, asian, american indian, pacific islander, other

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Harris, K.M., C.T. Halpern, E.A. Whitsel, J.M. Hussey, L. Killeya-Jones, J. Tabor, and S.C. Dean. 2019. Cohort Profile: The National Longitudinal Study of Adolescent to Adult Health (Add Health). International Journal of Epidemiology 48(5):1415-1425 https://doi.org/10.1093/ije/dyz115.

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